LifeMate: Agent Benchmark and Ablation

Train split · 1,545 samples · LFM3-1.7B and GIPformer 1.5-68M

Table 1: Cumulative ablation performance. Values are percentages except event counts. Higher is better (↑) or lower is better (↓). Best values are in bold.

	Contest	State		Behavior		Exploratory
Configuration	Contest Score ↑	SSR ↑	USMR ↓	SHR (Rec./Err.) ↑	MI/Abs F1 ↑	DDR ↓
Baseline	6.83	26.57	19.22	20.77 (279/1,343)	14.23	65.62
+ Enhanced Tool Descriptions	**7.72**	**27.30**	16.05	23.64 (265/1,121)	14.65	55.00
+ Enhanced System Prompt	7.66	27.09	**15.08**	23.83 (345/1,448)	14.88	11.63
+ Enhanced Sandbox Harness	7.64	26.85	16.31	**27.44** (309/1,126)	**14.93**	17.78
+ Sandbox 3.1 Runtime Checks	6.96	26.61	20.71	22.30 (188/843)	11.70	20.34
+ Sandbox 3.2 Runtime Checks	6.66	26.03	15.47	17.81 (135/758)	12.42	**8.33**

Contest Score: approximate reconstruction of the official specification, where fuzzy tokenization uses whitespace and may differ from the official scorer. SSR: sample-mean sub-goal state recall. USMR: sample-level unintended state mutation rate. SHR: self-healing percentage followed by recovered over total error events. MI/Abs F1: pooled missing-information and abstention F1. DDR uses 1,167 paired state-changing samples, where full/target matches are 33/96 (Baseline), 36/80 (+ Tool Desc.), 38/43 (+ Sys. Prompt), 37/45 (+ Sandbox), 47/59 (+ Sandbox 3.1), 33/36 (+ Sandbox 3.2).

Table 2: Outcome and execution diagnostics over 1,545 samples formatted as count (share). Counts across rows may overlap. Best values are in bold.

Category	Diagnostic	Baseline	+ Tool Desc.	+ Sys. Prompt	+ Sandbox	+ Sandbox 3.1	+ Sandbox 3.2
Outcome	Final state match	367 (23.75%)	386 (24.98%)	**389 (25.18%)**	380 (24.60%)	371 (24.01%)	373 (24.14%)
	Final state mismatch	1,178 (76.25%)	1,159 (75.02%)	**1,156 (74.82%)**	1,165 (75.40%)	1,174 (75.99%)	1,172 (75.86%)
	Exact ordered tool calls	87 (5.63%)	99 (6.41%)	**100 (6.47%)**	**100 (6.47%)**	90 (5.83%)	87 (5.63%)
	State match, gold has calls	270 (17.48%)	282 (18.25%)	**286 (18.51%)**	278 (17.99%)	273 (17.67%)	272 (17.61%)
	State match, no calls	215 (13.92%)	237 (15.34%)	226 (14.63%)	226 (14.63%)	203 (13.14%)	211 (13.66%)
Execution	No submitted tool calls	714 (46.21%)	755 (48.87%)	709 (45.89%)	709 (45.89%)	**673 (43.56%)**	701 (45.37%)
	No calls, gold has calls	639 (41.36%)	675 (43.69%)	630 (40.78%)	630 (40.78%)	**609 (39.42%)**	636 (41.17%)
	Reached 20-turn limit	95 (6.15%)	70 (4.53%)	89 (5.76%)	85 (5.50%)	62 (4.01%)	**56 (3.62%)**
	Repeated tool calls (attempted)	161 (10.42%)	**137 (8.87%)**	194 (12.56%)	183 (11.84%)	196 (12.69%)	148 (9.58%)
	Repeated tool calls (executed)	161 (10.42%)	137 (8.87%)	194 (12.56%)	182 (11.78%)	41 (2.65%)	**27 (1.75%)**
	Output parsing failure	12 (0.78%)	6 (0.39%)	4 (0.26%)	**0 (0.00%)**	**0 (0.00%)**	**0 (0.00%)**
	Hit context limit	**3 (0.19%)**	8 (0.52%)	9 (0.58%)	7 (0.45%)	6 (0.39%)	7 (0.45%)

All configurations use a 16,384-token context with no separate per-turn output cap and identical ASR transcripts. Attempted repeated calls include precheck-blocked calls, whereas executed repeated calls count only calls executed in ToolEnv.

Table 3: Observed language-model token usage across ablation stages.

Token measure	Configuration	Mean	Median	P90	P95	Maximum
First-turn input / sample	Baseline	4,505	4,374	6,400	6,954	8,487
	+ Tool Desc.	5,229	5,100	7,411	8,056	9,622
	+ Sys. Prompt	5,510	5,381	7,692	8,337	9,903
	+ Sandbox	5,510	5,381	7,692	8,337	9,903
	+ Sandbox 3.1	5,661	5,532	7,843	8,488	10,054
	+ Sandbox 3.2	5,583	5,454	7,765	8,410	9,976
Input summed over turns / sample	Baseline	26,125	7,021	85,876	140,433	258,719
	+ Tool Desc.	27,460	7,431	91,582	150,050	255,237
	+ Sys. Prompt	31,639	8,723	108,777	170,480	273,116
	+ Sandbox	30,498	8,723	99,937	155,400	286,976
	+ Sandbox 3.1	33,987	14,960	96,226	151,753	249,550
	+ Sandbox 3.2	26,677	10,097	71,621	137,776	279,365
Output / turn	Baseline	284	67	814	1,425	12,097
	+ Tool Desc.	302	93	855	1,464	6,555
	+ Sys. Prompt	285	52	921	1,478	13,077
	+ Sandbox	303	67	968	1,562	13,077
	+ Sandbox 3.1	290	97	849	1,424	9,479
	+ Sandbox 3.2	303	98	902	1,518	10,460
Output summed over turns / sample	Baseline	1,115	370	3,308	4,583	12,097
	+ Tool Desc.	1,084	328	3,359	4,664	8,634
	+ Sys. Prompt	1,138	359	3,385	4,460	13,077
	+ Sandbox	1,173	356	3,388	4,553	13,077
	+ Sandbox 3.1	1,307	568	3,613	4,831	9,479
	+ Sandbox 3.2	1,061	366	3,168	4,201	10,460

A turn is one generation, and input tokens include the prompt and conversation history at that turn. Summed input counts include shared prompt and history tokens across turns. Percentiles use the order statistic at index $\lfloor p(n-1) \rfloor$. Output counts include all generated text.